

Reviewer Pack — SOS Moment Matrix Spectral Gap and Max-CUT Approximation Rat...

Entry 49361c02c035 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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SOS Moment Matrix Spectral Gap and Max-CUT Approximation Ratio

Entry ID: 49361c02c035 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-21 08:09:37 UTC

1. Conjecture

Field A (mathematical branch): Real Algebraic Geometry **Field B** (complexity object): SOS Degree for Max-CUT

Statement:

For every Max-CUT instance with n variables, if the degree- d SOS moment matrix M_d has minimal eigenvalue $\lambda_{\min}(M_d) \geq \varepsilon$, then the integrality gap of the d -round SOS relaxation is $\leq 1 - \varepsilon$. Conversely, if the integrality gap exceeds 0.878, then $\lambda_{\min}(M_d) < 1 - 0.878$ for all $d \leq \log n$.

Rationale (proposer's reasoning):

The spectral gap of the moment matrix encodes the SOS relaxation's ability to capture low-degree certificates. By linking eigenvalue thresholds to approximation ratios, we expose structural constraints between algebraic geometry and optimization hardness, avoiding algebrization barriers through explicit spectral analysis.

Taxonomy category: SOS_DEGREE (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 735a0e68b05e8960

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	SAFE	SAFE

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.9s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_max_cut_instance(n):
        edges = set()
        for i in range(n):
            for j in range(i + 1, n):
                if random.random() < 0.5:
                    edges.add((i, j))
        return edges

    def degree_matrix(edges, n):
        deg = [0] * n
        for u, v in edges:
            deg[u] += 1
            deg[v] += 1
        D = [[deg[i] if i == j else 0 for j in range(n)] for i in range(n)]
        return D

    def laplacian_matrix(D):
        n = len(D)
        L = [[0] * n for _ in range(n)]
        for i in range(n):
            for j in range(n):
                if i == j:
                    L[i][j] = D[i][i]
                elif (i, j) in edges or (j, i) in edges:
                    L[i][j] = -D[i][j]
        return L

    def eigenvalues(matrix):
        n = len(matrix)
        if n == 1:
            return [matrix[0][0]]
```

```

# Gaussian elimination to transform matrix into upper triangular
for i in range(n):
    max_row = max(range(i, n), key=lambda x: abs(matrix[x][i]))
    matrix[i], matrix[max_row] = matrix[max_row], matrix[i]

    for j in range(i + 1, n):
        factor = matrix[j][i] / matrix[i][i]
        for k in range(n):
            matrix[j][k] -= factor * matrix[i][k]

# Extract eigenvalues from the diagonal
return [matrix[i][i] for i in range(n)]

def sos_moment_matrix(edges, n, d):
    D = degree_matrix(edges, n)
    L = laplacian_matrix(D)
    M_d = [[0] * n for _ in range(n)]

    for k in range(d + 1):
        for u in range(n):
            for v in range(u, n):
                if (u, v) in edges or (v, u) in edges:
                    M_d[u][v] += Fraction(1, 2 * d)
                    M_d[v][u] += Fraction(1, 2 * d)

    return M_d

def min_eigenvalue(matrix):
    eigenvals = eigenvalues(matrix)
    return min(eigenvals)

n = random.randint(5, 40)
edges = generate_max_cut_instance(n)
d_values = [2, 3, 4]
lambda_min = float('inf')
gap = 1.0

for d in d_values:
    M_d = sos_moment_matrix(edges, n, d)
    lambda_min_d = min_eigenvalue(M_d)
    if lambda_min_d < lambda_min:
        lambda_min = lambda_min_d

# Calculate integrality gap
integrality_gap = 1 - (lambda_min_d / (n * (n - 1) / 2))
if integrality_gap > gap:
    gap = integrality_gap

epsilon = lambda_min
conjecture_holds = gap <= 1 - epsilon and gap <= 0.878 + epsilon

return {
    "metric_name": "Integrality Gap",
    "metric_value": gap,
    "instances_tested": len(d_values),

```

```

        "conjecture_holds": conjecture_holds,
        "counterexample": "" if conjecture_holds else f"Gap {gap} exceeds 1 - epsilon={1 - epsilon}"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_gap = sum(r["metric_value"] for r in results) / len(results)
    std_gap = math.sqrt(sum((r["metric_value"] - mean_gap) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_gap} std={std_gap} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"Gap exceeds 1 - epsilon\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE reason=insufficient_evidence")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

value': 1.0142857142857142, 'instances_tested': 3, 'conjecture_holds': False, 'counterexample': 'Gap
epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0714285714285714, 'instances_tested': 3,
epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0059751773049646, 'instances_tested': 3,
epsilon=525/188'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0130870210295446, 'instances_tested': 3,
epsilon=23245/5392'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0416666666666667, 'instances_tested': 3,
epsilon=5/2'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.004379479878373, 'instances_tested': 3,
epsilon=131545706/32251283'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.01875, 'instances_tested': 3, 'conjecture_holds': False,
epsilon=13/4'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0042481778622288, 'instances_tested': 3,
epsilon=16057508/4367783'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0113157894736842, 'instances_tested': 3,
epsilon=587/200'}
TRIAL: {'metric_name': 'Integrality Gap', 'metric_value': 1.0110294117647058, 'instances_tested': 3,
epsilon=5/2'}
RESULT: FALSIFIED counterexample="Gap exceeds 1 - epsilon" first_failing_seed=2

```

8. Critic adversarial review

Critic verdict: CHALLENGE

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

parse_fail | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	103330
2	preregistration	ollama_remote	qwen3:8b	0	0	31767
3	novelty	ollama_remote	qwen3:8b	0	0	27541
4	novelty	ollama_remote	qwen3:8b	0	0	20875
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18121
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13721
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13063
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13222
9	critic	ollama_remote	qwen3:8b	0	0	38867
10	judge	ollama_remote	qwen3:8b	0	0	28331

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 308837 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/49361c02c035.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/49361c02c035.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/49361c02c035.tar.gz (if generated)